Western Mathematics

TOPIC REVISION
QUESTIONS

Mathematics Standard 2

MS-M7: Rates and Ratios

Section I –Multiple Choice Questions (1 Mark each)

Section II – Longer Questions (Marks indicated beside questions)

Section I

16 marks

1. From a group of 240 students, 100 are at sport and the rest are in class. What is the ratio in simplest terms of those at sport to those in class?

   (A) 12 : 5

   (B) 5 : 7

   (C) 17 : 5

   (D) 7 : 5

2. The ratio of flour to sugar in a recipe is 8 : 3.

   If I have 20 cups of flour, how much sugar should I use?

   (A) 7.5 cups

   (B) 8 cups

   (C) 9.5 cups

   (D) 10 cups

3. The rate of flow of water from a dam is 24 kilolitres/hour. How long would it take for 150 kilolitres to flow from the dam?

   (A) 6 hours and 15 minutes

   (B) 6 hours and 25 minutes

   (C) 8 hours

   (D) 8 hours and 15 minutes

4. The directions to mix fertiliser say to add 4 teaspoons of granules to 500 millilitres of water. If each teaspoon holds 4 grams of granules, what is the rate for mixing the fertiliser?

   (A) 8 grams/litre

   (B) 16 grams/litre

   (C) 32 grams/litre

   (D) 64 grams/litre

5. Mike runs 20 laps of the oval per day. If each lap is 500 metres, how many kilometres would he have run in the month of January?

(A) 280 kilometres

(B) 300 kilometres

(C) 310 kilometres

(D) 620 kilometres

6. When we divide 2 hours in the ratio 7 : 5, what are the resulting amounts?

(A) 70 min : 50 min

(B) 50 min : 70 min

(C) 35 min : 25 min

(D) 14 min : 10 min

7. A patient is given medication by way of drops delivered through a tube. One millilitre of medication contains 10 drops. If the patient needs 2.16 litres of the medication per day, how many drops per minute should the delivered?

(A) 1.5 drops.

(B) 6 drops.

(C) 9 drops.

(D) 15 drops.

8. What percentage of \$45·00 is \$1·34? Answer correct to 1 decimal place.

(A) 2.9%

(B) 3.0%

(C) 13.4%

(D) 43.7%

9. A map shows its scale as 1 cm : 500 km.

Which ratio this equivalent to?

(A) 1 : 5 000

(B) 1 : 5 000 000

(C) 1 : 50 000 000

(D) 1 : 5 000 000 000

10. Mary's car uses 20 litres of fuel to travel 250 km. The consumption rate is:

(A) 8 litres per 100 kilometres

(B) 12.5 litres per 100 kilometres

(C) 16 litres per 100 kilometres

(D) 25 litres per 100 kilometres

11. Misty and Britni plan to divide their earnings from a fundraising drive between their two favourite charities in the ratio 4 : 5.

They raised $450 altogether. What is the difference between the amounts given to the two charities?

(A) $9

(B) $50

(C) $200

(D) $250

12. Marty calculates that his bicycle travels at 5 metres per second. What is this speed in kilometres per hour?

(A) 0.3 km/h

(B) 1.4 km/h

(C) 18 km/h

(D) 180 km/h

13. Tarax brand of soft drink is available in the following packaging. Which is the best value?

(A) 2 litre bottle for $2.50.

(B) 1.25 litre bottle for $1.50

(C) 600 ml bottle for $1.20

(D) 4 pack of 300 ml bottles for $2.20

14. In carrying out a survey of wombats in a forest, Kerrie traps 15 wombats, tags them and releases them. A week later she captures 12 wombats in the same forest and finds that 3 of them are tagged.
How many wombats would she estimate there are in the forest?

(A) 15

(B) 20

(C) 45

(D) 60

15. To estimate the total number of feral cats in a national park, rangers caught 25, tagged them and then released them.

A few days later they caught 20 cats and noted that 7 of these were tagged.

If we use x to represent the total number of feral cats, which is the correct calculation to estimate the value of x?

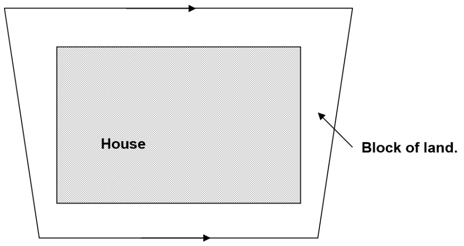
(A) $x = \frac{7 \times 20}{25}$

(B) $x = \frac{25}{7 \times 20}$

(C) $x = \frac{25 \times 7}{20}$

(D) $x = \frac{25 \times 20}{7}$

16. A scale drawing of a house, showing its position on the block of land on which it is to be built is shown below. The scale is 500:1.



What area of the block is not covered by the house (rounded to the nearest 10 m^2)?

(A) 112 m^2

(B) 130 m^2

(C) 210 m^2

(D) 350 m^2

Western Mathematics
Mathematics Standard 2
MS-M7: Rates and Ratios
TOPIC REVISION QUESTIONS - Section II

Instructions
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.

Question 17 (2 marks) **Marks**

A cylindrical water tank has a diameter of 3.6 metres and a height of 2.2 metres.

(a) Find the volume of the tank in cubic metres. **1**

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(b) If the tank is filled at a rate of 400L/h, how many hours would it take to fill the tank? (Answer to the nearest hour.) **1**
1 m^3 holds 1000 litres.

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Question 18 (2 marks)

A particular concrete mix uses cement : sand : gravel in the ratio 3 : 5 : 4 by volume.

(a) Todd needs to make 36 buckets of concrete mix, How much cement, sand and gravel will he need? **1**

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(b) Kerry has enough cement to fill 4.5 buckets. How many buckets of sand and gravel will he need to mix with this to make as much cement as possible? **1**

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Question 19 (3 marks)

(a) A car uses fuel at the rate of 9 kilometres/litre. How far could it travel on 40 litres of fuel? **1**

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(b) Express the rate of 9 kilometres/litre as litres/100 km. **2**

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Question 20 (3 marks)

Jo conducts a survey of the entire school population, recording the method of transport that students use to get to school. The results are shown below.

Type of transport	Number of Students
Walk	120
Bicycle	150
Bus	350
Car	240
Other	140

(a) What is the ratio of those who travel by bus to those who travel by car, in simplest form? **1**

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(b) If a new farm brings an extra 14 students to the school who all travel by bus, what is the percentage increase in students who travel by bus? **2**

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Question 21 (5 marks)

Peter, who is 24, checks his resting heart rate by counting 17 beats in 20 seconds.,

(a) Express this in beats per minute. **1**

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(b) During exercise it increased to 28 beats in 15 seconds. Compare this to his resting rate. **2**

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(c) He wants to ensure his exercise rate does not go over the recommended maximum heart rate (MHR) which is calculated by subtracting his age from 220. **2**
How many beats should he count every 15 seconds if he was at his MHR?

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Question 22 (3 marks)

Adam works on a farm and needs to distribute fertiliser over two paddocks.

(a) The first paddock is 2 500 Ha and Adam uses a fertiliser called Manzec which requires a rate of 6.4kg/Ha. How many tonnes of Manzec are required? **1**

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(b) The second paddock is 3 600 Ha and Adam has 9 tonnes of a fertiliser called Zemank left from last year. How many kg/Ha would he be able to distribute of the leftover Zemank? **2**

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Question 23 (3 marks)

Cleen n Soft laundry powder is sold in three different sizes. **3**

- 750g for \$3.30
- 1kg for \$4.30
- 1.25kg for \$5.00

Express each of these as a price per kilogram and determine which is the best value.

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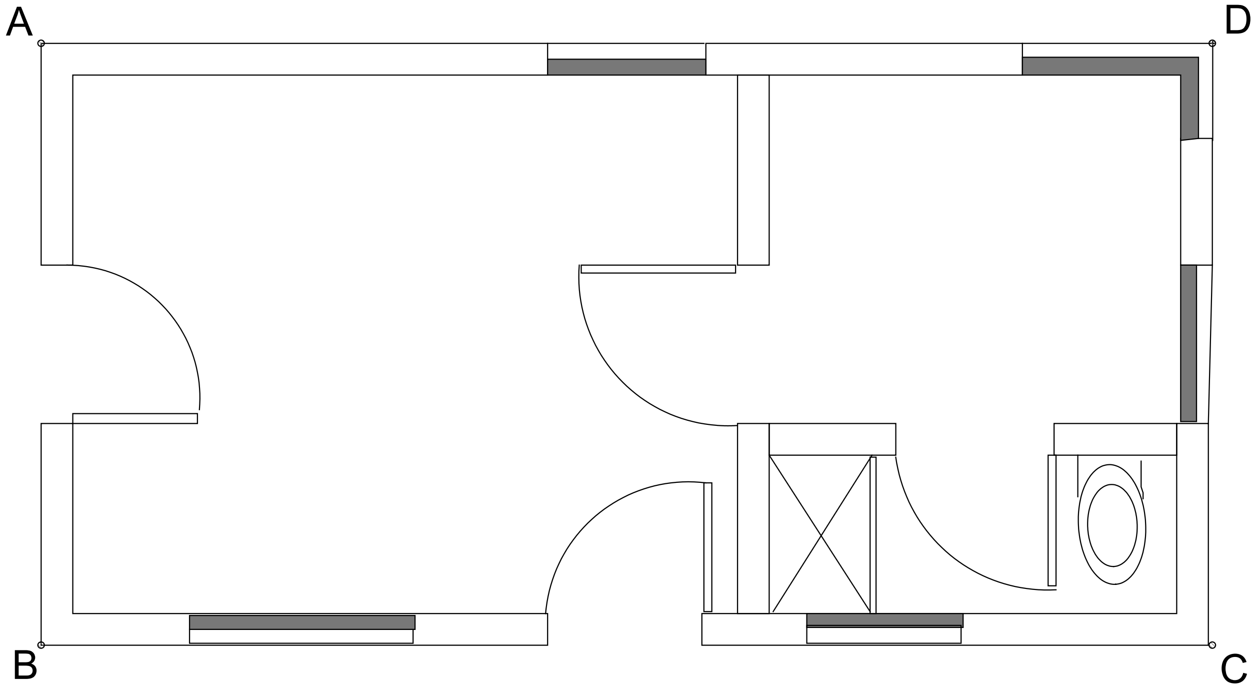
Question 24 (4 marks)

The floor plan of a holiday cabin is shown below.



(a) The width of the cabin, AB is 3 600mm. What is the scale of the drawing? **1**

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(b) What is the length of the cabin, BC (in millimetres to 2 sig figures)? **1**

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(c) What is the total floor area of the cabin, ABCD (in square metres to 2 sig figures)? **2**

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Question 25 (2 marks)

A park ranger catches and tags 64 rosellas in the Blue Mountains National Park. A week later she catches another 40 rosellas in the same park and finds that 15 of them are tagged. Estimate the number of rosellas in the park. **2**

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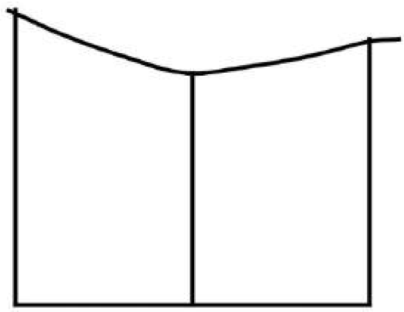
Question 26 (6 marks)

A scale drawing of a block of land is shown below. It has been divided into two section by an offset line.



Scale 1 : 2500

(a) Calculate the length on the ground of all straight lines shown above and write these on the diagram. **2**

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(b) Use two applications of the trapezoidal rule to find the area of the block. **2**

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(c) If 12 mm of rain fell on the block over a day. **2**
How many litres of water fell on the block? (1 m^3 holds 1000 litres)

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Question 27 (6 marks)

Andy's air conditioner has a rating of 4.0 kW when cooling and 5.5 kW when heating.

He pays 27c per kWh for electricity.

(a) He uses it for an average of 8 hours a day in summer for cooling. What would it cost him for the three months of summer? (December-February in a non-leap year.) **2**

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(b) He uses it for an average of 6 hours a day in winter for heating. What would it cost him for the three months of winter? (June- August.) **2**

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(c) A gas heater uses 0.25 litres of gas per hour to produce the same level of heating as the air-conditioner. If gas costs 80 cents/litre, would he be better off using gas for heating in winter? **2**

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Question 28 (4 marks)

(a) Justin travels a distance of 280 km, taking three and a half hours. **2**
If he maintains this speed, how many hours would a trip of 700 km take?

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(b) Kristina buys 24 games for her three children and divides them in the ratio of their ages, **2**
which are 9 years, 6 years and 3 years. How many games does each child receive?

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Question 29 (3 marks)

The fuel consumption of a car, in highway driving, is 12 km/L.

(a) How far (on a highway) could the car travel on a tank which holds 40 litres. **1**

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(b) Express the fuel consumption in L/100km. **2**

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Question 30 (2 marks)

On a scale drawing a square lawn measures 4cm wide, with a circular flower bed of diameter 2.4 cm at it's centre . The width of the real square lawn is 10 metres.

(a) What is the scale of the plan? **1**

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(b) What is the diameter of the real flower bed? **1**

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Question 31 (4 marks)

A plane uses 5 litres of fuel per second during take-off.

(a) Express this in kilolitres per hour. **2**

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(b) If the plane uses fuel at a 60% lower rate when cruising at altitude, how much would it use when cruising for 20 minutes? **2**

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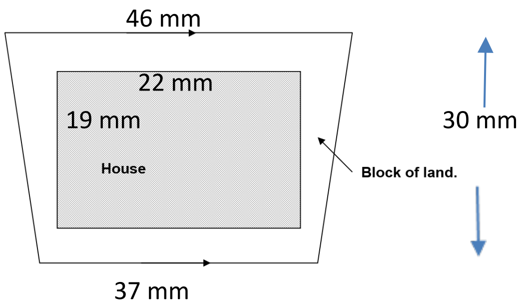
End of Revision

Western Mathematics MS-M7: Rates and Ratios Revision

Section I - Worked Solutions

No	Working	Answer
1	Those in class $= 240 - 100 = 140$ Sport : Class $= 100 : 140 = 5 : 7$	B
2	Flour : Sugar $= 8 : 3 = 20 : x$ Factor $= \dfrac{20}{8} = 2.5$ Sugar $= 3 \times 2.5 = 7.5$ cups	A
3	Rate $= 24$ kL/h No hours $= 150/24 = 6.25$ hours $=$ 6 hours and 15 min	A
4	Mass in 4 teaspoons $= 4 \times 4g = 16\ g$ Rate $= 16$ g/500 ml $= 16 \times 2g/500 \times 2\ ml$ $=$ 32g / 1000 ml $=$ 32g /litre	B
5	Distance run per day $= 20 \times 500$ $= 10\ 000$ m $= 10$ km Dist run in Jan $= 10 \times 31$ days $= 310$ km	C
6	Ratio of an hour $= 7 : 5$ So divided into $7 + 5 = 12$ parts Each part $= 120/12 = 10$ min Ratio $= 7 \times 10 : 5 \times 10$ $=$ 70 min : 50 min	A
7	Number of drops per day $= 2.16 \times 1000$ ml $\times 10$ $= 21\ 600$ drops per day Number of drops per min $= \dfrac{21600}{24 \times 60}$ $= 15$ drops per min	D
8	Percentage $= \dfrac{1.34}{45.00} \times 100$ $= 2.97777778$ $= 3.0\%$ (1 dec place)	B
9	1 cm : 500 km $=$ 1 cm : 500 000 m $=$ 1 cm : 50 000 000 cm $= 1 : 50\ 000\ 000$	C
10	Rate $= \dfrac{20}{250} \times 100$ litres per 100 km $= 8$ litres per 100 km	A
11	Misty : Britni $= 4 : 5$ So divided into $4 + 5 = 9$ parts	B

	Difference $= \$50$	
12	Speed $= 5$ m/s $= 5 \times 60 \times 60 = 18000$ m/hour $= \dfrac{18000}{1000} = 18$ km/hour	**C**
13	$A.$ $\$2.50 \div 2 = \$1.25$ per litre $B.$ $\$1.50 \div 1.25 = \$1.20$ per litre $C.$ $\$1.20 \div 0.6 = \$2.00$ per litre $D.$ $\$2.20 \div (4 \times 0.3) = \$1.83$ per litre	**B**
14	From second capture $\dfrac{3}{12}$ of wombats are tagged $= 25\%$ So 15 wombats is 25% so total wombats $= 15 \times 4 = 60$ wombats Alternative Let x = total number wombats $\frac{x}{15} = \frac{12}{3}$ $x = \frac{15 \times 12}{3}$ $= 60$	**D**
15	Let x = total number cats $\dfrac{x}{25} = \dfrac{20}{7}$ $x = \dfrac{25 \times 20}{7}$	**D**
16	Multiply measured distances by 500 and convert to metres.  46 mm $\Rightarrow$ 23000 mm $=$ 23.0 m 22 mm $\Rightarrow$ 11000 mm $=$ 11.0 m 37 mm $\Rightarrow$ 18500 mm $=$ 18.5 m 19 mm $\Rightarrow$ 9500 mm $=$ 9.5 m 30 mm $\Rightarrow$ 15000 mm $=$ 15.0 m Area land $= \frac{h}{2}(a+b)$ $= \dfrac{15}{2}(23 + 18.5) = 311.25 \text{ m}^2$ Area house $=$ lb $= 11 \times 9.5 = 104.5 \text{ m}^2$ Area not covered $= 311.25 - 104.5$ $= 206.75 \text{ m}^2$ $= 210 \text{ m}^2$ (Nearest 10 m^2)	**C**

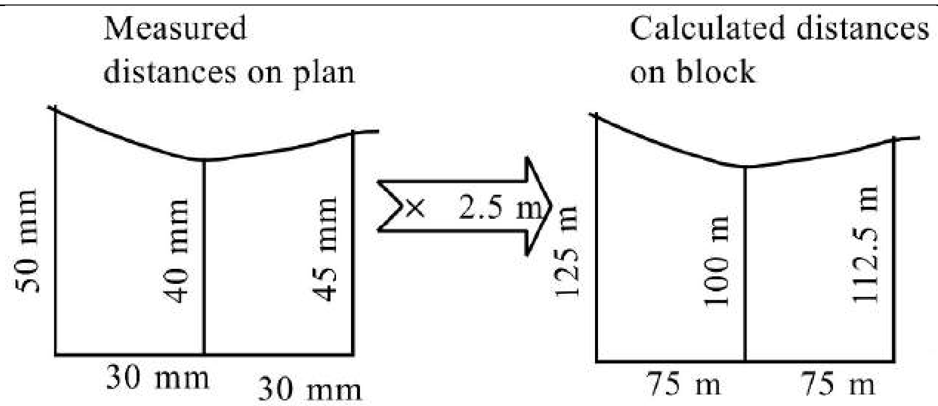
	MS-M7: Rates and Ratios Section II Solution	Marks
Question 17		
(a)	$V = \pi r^2 h$ i) $= \pi \times 1.8^2 \times 2.2$ $= 22.4 \text{ m}^3$	1
(b)	ii) Capacity of tank $= 22.4 \times 1000 = 22\ 400$ Litres Time taken $= 22\ 393 \div 400$ $= 55.98$ $= 56$ hours (nearest hour)	1
Question 18		
(a)	Total Parts in Concrete $= 3 + 5 + 4 = 12$ 36 buckets $= 3 \times 12 = 3$ lots of mix Amounts are Cement $= 3 \times 3 = 9$ buckets Sand $= 3 \times 5 = 15$ buckets Gravel $= 3 \times 4 = 12$ buckets	1
(b)	Cement $= 3$ parts of mix $4.5 \div 3 = 1 \cdot 5$ lots of mix Sand $= 1 \cdot 5 \times 5 = 7.5$ buckets Gravel $= 1 \cdot 5 \times 4 = 6$ buckets	1
Question 19		
(a)	Number of km $= 9 \times 40 = 360$ km	1
(b)	9 km/litre $=$ 1 litre/9 km 1 litre/9 km = 100 litres/900km $= \dfrac{100}{9}$ litres/100km $=$ 11 litres/100km (nearest litre)	2
Question 20		
(a)	(i) Ratio $= 350 : 240 = 35 : 24$	1
(b)	(ii) Percentage increase $= \dfrac{14}{350} \times 100$ $= 4\%$ increase	2

	MS-M7: Rates and Ratios Section II Solution	Marks
Question 21		
(a)	beats/min $=$ beats/60 seconds 17 beats/20 seconds $=$ 17×3 beats/60 seconds $(3\times 20 \text{ sec} = 60 \text{ sec})$ Resting Heart Rate $=$ 51 beats/minute	**1**
(b)	28 beats/15 seconds $=$ 28×4 beats/60 seconds $=$ 112 beats/minute This is more than double his resting heart rate	**2**
(c)	MHR $= 220 - 24 = 196$ beats/minute 196 beats/60 seconds $= \frac{196}{4}$ beats/15 seconds $=$ 49 beats/15 seconds at MHR	**2**
Question 22		
(a)	(i) Amount $= 2\,500 \times 6.4$ $= 16000$ kg $= 16$ tonnes	1
(b)	(ii) Rate in kh/Ha $= 9000 \div 3600$ $= 2.5$ kg/Ha	2
Question 23		
	$*$ \$3.30 per 750 g $=$ \$3.30 per 0.75 kg $= 3.30 \div 0.75$ $=$ \$4.40 per kg $*$ \$4.30 per 1 kg $=$ \$4.30 per kg $*$ \$5.00 per 1.25 kg $= 5.00 \div 1.25$ $=$ \$4.00 per kg 1.25 kg is best value	3
Question 24		
(a)	Scale = 80 : 3600 = 1 : 45	1
(b)	Length = 154 $\times$ 45 = 6930 = 6900 mm (2 sf)	1

<table>
<tr><td></td><td>MS-M7: Rates and Ratios

Section II Solution</td><td>Marks</td></tr>
<tr><td>(c)</td><td>$$\begin{aligned} Area &= 3.6 \times 6.93 \\ &= 24.946 \\ &= 25\ m^2 \ (2\text{ sf}) \end{aligned}$$</td><td>2</td></tr>
<tr><td colspan="2">Question 25</td><td></td></tr>
<tr><td>(a)</td><td>Percentage tagged = $\frac{15}{40} \times 100 = 37.5\%$

37.5% of rosellas = 64 birds.

Total rosellas = 64 $\div 37.5 \times 100 = 171$ rosellas</td><td>2</td></tr>
<tr><td colspan="2">Question 26</td><td></td></tr>
<tr><td>(a)</td><td>

1 mm × 2500 ÷ 1000 = 2.5
× 2500 for scale
÷ 1000 to convert to metres
"so scale is the same as
1 mm : 2.5m</td><td>2</td></tr>
<tr><td>(b)</td><td>$\text{Area}_L = \frac{75}{2}(125 + 100) = 8437 \cdot 5$
$\text{Area}_R = \frac{75}{2}(100 + 112 \cdot 5) = 7968.75$
$\text{Area Block} = 8437 \cdot 5 + 7968 \cdot 75 = 16406 \cdot 25\ \text{m}^2$</td><td>2</td></tr>
</table>

	MS-M7: Rates and Ratios Section II Solution	Marks
(c)	$\text{Depth} = 12\text{ mm} = 0 \cdot 012\text{ m}$ $\text{Volume} = A\,h$ $= 16406 \cdot 25 \times 0 \cdot 012$ $= 196 \cdot 875\text{ m}^3$ $\text{Amount (litres)} = 196 \cdot 875 \times 1000$ $= 196\,875\text{ litres}$	2
Question 27		
(a)	$\text{Usage per day(Summer)} = 8 \times 4 = 32\text{ kWh}$ $\text{Cost per day(Summer)} = 32 \times 0 \cdot 27 = \$8 \cdot 64$ $\text{Days in summer} = 31 + 31 + 28 = 90\text{ days}$ $\text{Cost (Summer)} = 90 \times 8 \cdot 64 = \$777 \cdot 60$	2
(b)	$\text{Usage per day (Winter)} = 6 \times 5 \cdot 5 = 33\text{ kWh}$ $\text{Cost per day(Winter)} = 33 \times 0 \cdot 27 = \$8 \cdot 91$ $\text{Days in winter} = 30 + 31 + 31 = 92\text{ days}$ $\text{Cost (Winter)} = 92 \times 8 \cdot 91 = \$819 \cdot 72$	2
(c)	$\text{Gas usage per day (Winter)} = 6 \times 1 \cdot 35 = 8 \cdot 1\text{ litres}$ $\text{Gas cost per day(Winter)} = 8 \cdot 1 \times 0 \cdot 85 = \$6 \cdot 885 = \$6 \cdot 89\text{ per day}$ Since air cond was $\$8 \cdot 91$ per day, the gas is cheaper.	2
Question 28		
(a)	$S = \frac{D}{T}$ $= \frac{280}{3 \cdot 5} = 80\text{ km/h}$ $T = \frac{D}{S}$ $= \frac{700}{80} = 8 \cdot 75$ $= 8\text{ hours and }45\text{ min}$	2

	MS-M7: Rates and Ratios Section II Solution	Marks
(b)	Ratio $= 9 : 6 : 3$ $= 3 : 2 : 1$ (= 6 parts) Divide 24 in ratio $3 : 2 : 1$ Each part = $24 \div 6 = 4$ Nine year old gets $3 \times 4 = 12$ Six year old gets $2 \times 4 = 8$ Three year old gets $1 \times 4 = 4$	2

Question 29		
(a)	Car travels 12km/L, so on 40 L can travel $40\times 12 = 480$ km	1
(b)	$12 \text{ km/L} = \frac{1}{12} \text{ L/km}$ $= 0.08\dot{3} \text{ L/km}$ $= 8.3 \text{ L/100km}$ (1 dec place)	2
Question 30		
(a)	Scale is 4 cm : 10 metres = 1 cm : 2.5 m = 1 : 250	1
(b)	Diameter on drawing is 2.4 cm Diameter of real bed = 2.4×2.5 = 6 metres.	1
Question 31		
a)	5 litres per second $= 5\times 60\times 60$ litres per hour $=$ 18000 litres per hour $= \frac{18000}{1000}$ $=$ 18 kilolitres per hour	2
b)	60% reduction $=$ 40% of 18 kL / hour $= 0.4 \times 18 = 7.2$ kL / hour Fuel used in 20 min $=$ Fuel used in $\frac{1}{3}$ of an hour $= \frac{1}{3} \times 7.2$ $=$ 2.4 kilolitres	2